```
Test Set Performance:
Accuracy: 0.9649
Precision: 0.9589
Recall: 0.9859
F1 Score: 0.9722
```

Summary of Each Measure

- Accuracy : It measures the proportion of correctly classified instances out of all instances.
- Precision : It's the ratio of true positive predictions to the total predicted positives. It indicates how many of the predicted positive instances are actually positive.
- Recall : Recall, also known as sensitivity, is the ratio of true positive predictions to the total actual positives. It shows how many of the actual positive instances were correctly identified.
- F1-score : The F1-score is the harmonic mean of precision and recall, providing a balanced measure of both. It is particularly useful when there is an uneven class distribution.

- Accuracy (0.949) : The accuracy score indicates that the model correctly classifies approximately 94.9% of the instances in the test set.
  This is a strong overall performance, suggesting that the model is effective at making correct predictions.
- Precision (0.9589) : The precision score shows that about 95.89% of the predicted positive instances are actually positive.
- This high precision indicates that the model has a low rate of false positives, meaning it rarely misclassifies negative instances as positive.
- Recall (0.9859) : The recall score reveals that the model successfully identifies 98.59% of the actual positive instances.
  This excellent recall suggests that the model is very effective at detecting true positive cases, with a low rate of false negatives.
- F1-score (0.9722) : The F1-score, which is the harmonic mean of precision and recall, is 0.9722.
  This high F1-score indicates a well-balanced performance between precision and recall, confirming that the model not only makes accurate positive predictions but also effectively identifies most of the true positive cases.